



Midterm Assignment

Sehyun Kwon · Due October 23, 2026, via LMS

This assignment replaces the midterm exam. You will use AI to work on an open mathematical problem and write a report on your attempts. In this course, the aim is to learn how to use AI effectively, check its answers, and revise an approach when an argument fails.

1 Background

AI systems have recently been used to find proofs and counterexamples to open problems. Research announcements in 2026 include:

- OpenAI's reported results on finite-time blowup for forced Navier–Stokes and unforced three-dimensional Euler equations, and its disproof of the Erdős unit-distance conjecture.
- Work by Axiom Math and collaborators on the Chen–Gendron conjecture, Fel's numerical semi-group conjecture, and Stanley and collaborators' alternating-permutation problem.
- Google DeepMind and collaborators' work on Erdős problems 652 and 1051.

These are research reports, and their independent verification and acceptance vary. In particular, the reported forced Navier–Stokes result does not settle the unforced case. The links above lead to the reports.

2 The problem

The Erdős–Straus conjecture asks: for every integer $n \geq 2$, do there exist positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}?$$

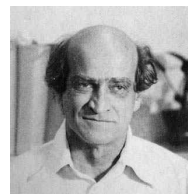
The integers x, y, z **need not be distinct**. For example, when $n = 5$, $\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$.

The conjecture remains open in the recent research literature. Your task is to investigate this statement, not a version requiring distinct denominators.

To prove it true, give a proof that works for **every** $n \geq 2$. To prove it false, exhibit a particular $n \geq 2$ and prove that **no** positive integer triple (x, y, z) works. Checking many values, or failing to find a solution within a limited search range, is not enough.



Paul Erdős
1913–1996



Ernst Gabor Straus
1922–1983

Photographs and biographical dates: MacTutor History of Mathematics (Erdős; Straus).

3 How to approach the assignment

Use what we have learned about logic, proofs, and induction to check the arguments you obtain. You can ask AI to break the problem into smaller questions, try different approaches, or look for errors in a proposed proof. When an attempt fails, explain the problem to the AI and revise your approach.

Do not accept an answer simply because AI calls it a proof. Check each step: does the conclusion follow from the assumptions? Is an assumption missing? Have all cases been covered? Check revised answers just as carefully as the first one.

You may use multiple AI agents to propose proofs, search for counterexamples, and review each other's work. Their agreement does not establish correctness. Lean is also optional. If you use it, make sure the formal statement matches the problem and that no unproved gaps or extra assumptions supply the result you are trying to prove.

4 The report

The first line of your report must state one of the following:

- (a) **Solution status:** Complete solution claimed
- (b) **Solution status:** Partial solution only
- (c) **Solution status:** No solution established

If you claim a partial solution, state what you proved and what remains unresolved immediately after this line.

Include the following in your report:

1. Your main prompts and the corresponding AI responses.
2. The approaches you tried and the tasks you gave to AI.
3. Specific claims you checked, how you checked them, and why you accepted or rejected them.
4. The results you established and the steps where you got stuck.
5. How you changed your prompts or approach after a failed attempt, and what happened next.
6. Your final assessment, supported by the relevant proofs. For a complete solution, give the full proof or a rigorously justified counterexample. For a partial result, prove it and explain why it is nontrivial.

A conversation log alone is not enough. Explain what you found, what went wrong, and how this affected your next attempt.

5 Grading

The assignment is graded out of **100 points**. A base score of **0–50 points** is awarded for how thoroughly and specifically you explain your work. The following adjustments apply:

- **+50 points:** you claim a complete solution and the proof is correct.
- **–10 points:** you claim a complete solution but the proof is incorrect.
- **+10 points:** you do not solve the full problem, but prove a partial result and convincingly explain why it is nontrivial.
- **No adjustment:** you report no complete solution and do not meet the partial-result criterion.

Thus, careful work can earn up to 50 points without a solution, or up to 60 with a qualifying partial result. A proof of only some cases is not a complete solution. Be clear about what your argument actually establishes.